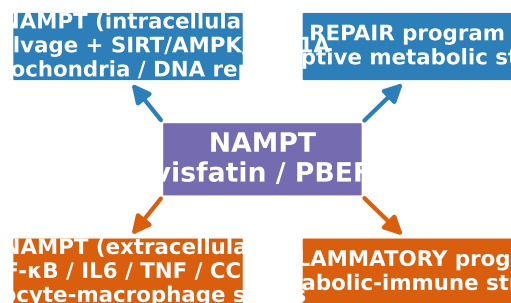


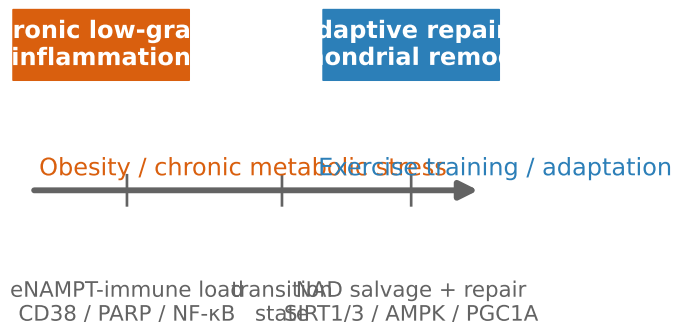
State-dependent NAMPT-NAD inflammatory-repair axis in obesity and exercise adaptation

a NAMPT dual identity



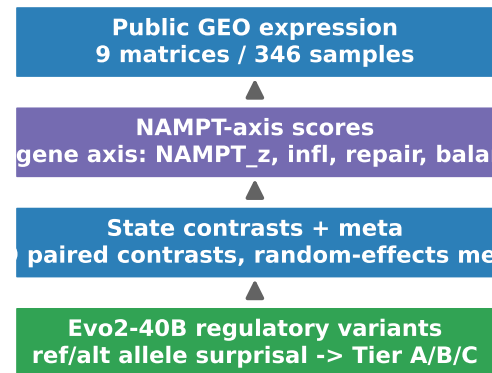
Key question: when does NAMPT signal repair vs. inflammatory load?

b State-dependent axis



The same gene set, different states
-> balance_score flips direction

c Multi-layer public-data evidence



d Repair program (transcript evidence)

co-expression support:

GSE272133 repair modules $\alpha = 0.79-0.91$

GSE305038 α mean 0.78; GSE32575 α mean 0.61

NAMPT NMNAT1/2/3 NAPRT NADSYN1
SIRT1 SIRT3 SIRT6 PRKAA1/2
PPARGC1A TFAM NRF1 SOD2
CAT GPX1 ATG5 BECN1 XRCC1

e Inflammatory program (transcript evidence)

state evidence:

acute exercise: NAMPT_z +0.85 (92% agree)

balance -0.62 (83%); NF-kB-module sensitive

NAMPT CD38 BST1 PARP1/2
NFKB1 RELA TNF IL6 IL1B
CCL2 CXCL8 TLR4 NLRP3
CASP1 CD14 CD68 ITGAM

Acute exercise (k=3) 0.12

Obesity (k=3) 0.10

Cell model (k=2) 0.07

6-wk training (n=3) 0.27

repair-negative <- balance -> repair-positive